

Trade Finance Analysis

Mexico, Peru, Chile, and Brazil (2010-2025)

Country Analysis Pipeline

November 2025

Data Characteristics Summary

Characteristic	Mexico	Peru	Chile	Brazil
Bank-level data	✓	✓	✓	×
Firm size breakdown	×	✓	×	✓
Economic sector	×	×	×	✓
Geographic region	×	×	×	✓
Total liabilities	✓	×	✓	×
Instrument type (L/C)	✓	×	✓	×
Period	2022-2025	2010-2024	2022-2024*	2012-2024
Observations	2,204	13,785	28,729	838,166

Key Notes:

- **Chile*:** Full series 2015-2024 available, but analysis uses 2022-2024 only (CMF period) due to accounting system break
- **Brazil:** No bank-level data in public SCR; compensated with detailed borrower characteristics
- **Trade data:** Global Macroeconomic Database (GMD) used for exports/imports across all countries

① Data Sources and Methodology (7 slides)

- Overview and comparability
- Country-specific data characteristics
- GMD integration

② Mexico Analysis (6 graphs)

③ Peru Analysis (7 graphs)

④ Chile Analysis (9 graphs)

⑤ Brazil Analysis (8 graphs)

Total: 30 graphs across 4 countries

Data Sources Overview

Country	Period	Source	Coverage
Mexico	2022-2025	CNBV Reporte A-1219	Letters of Credit (account 202401504003)
Peru	2010-2024	SBS Balance de Comprobación	Foreign-trade credit portfolio
Chile	2015-2024	CMF API (Bancos)	Trade finance accounts (29 accounts CMF period)
Brazil	2012-2024	BCB Monthly Reports	Foreign-trade credit (aggregate only)

Trade Data Source: Global Macroeconomic Database (GMD) for annual exports and imports across all four countries.

Mexico: Data Characteristics

Source: CNBV Reporte A-1219

Coverage:

- Account 202401504003: Import Letters of Credit $\leq$ 180 days
- Monthly frequency: January 2022 - August 2025
- Bank-level data: 53 institutions
- Includes total liabilities for percentage calculations

Limitations:

- Only captures short-term import L/C
- Export L/C and other trade instruments not included
- Period: 2022-2025 only

Source: SBS Balance de Comprobación (Resolución SBS N° 11356-2008)

Coverage:

- "Comercio Exterior" credit portfolio
- Monthly frequency: October 2010 - December 2024
- Bank-level data: 19 institutions
- Borrower characteristics: firm size, sector, region

Limitations:

- Does not separate by instrument type
- Firm size data incomplete for some periods
- Aggregate "foreign-trade credit" category

Chile: Data Characteristics and Accounting Break

Source: CMF API - Información Financiera de Bancos

Coverage:

- 2015-2021: SBIF accounting (11 trade finance accounts identified)
- 2022-2024: CMF/IFRS 9 accounting (29 accounts identified)
- Monthly frequency, bank-level data: 27 institutions
- Includes total liabilities (account 2000000000) for CMF period

Accounting Transition:

- **January 2022: Structural break** (SBIF → CMF/IFRS 9 transition)
- 0% account code overlap between periods
- **Analysis uses CMF period only (2022-2024)** for consistency
- Jump in 2022 reflects account identification change (11 → 29 accounts)

Brazil: Data Characteristics

Source: BCB Monthly Reports

Coverage:

- "Comércio Exterior" credit operations
- Monthly frequency: January 2012 - December 2024
- 838,166 observations
- Borrower characteristics: firm size, sector, geographic region

Limitations:

- No bank-level data (publicly available data aggregated only)
- Cannot analyze bank concentration or top institutions
- Does not separate by instrument type

Global Macroeconomic Database (GMD) Integration

Purpose: Calculate trade finance intensity metrics

Variables Used:

- Annual exports (USD millions)
- Annual imports (USD millions)
- Total trade = exports + imports

Data Source Change:

- **Previous:** BACI/CEPII bilateral trade database
- **Current:** Global Macroeconomic Database (GMD)
- **Reason:** BACI/CEPII latest available data only extends through 2023. GMD provides data through 2024-2025.

Methodology:

- Monthly TF stock divided by annual trade flows
- Enables cross-country comparisons of TF penetration
- Example: $\text{tf_pct_exports} = (\text{monthly_tf_stock} / \text{annual_exports}) \times 100$

Data Comparability Across Countries

Comparable Metrics:

- TF as % of exports (intensity measure)
- TF as % of total trade (broader penetration)
- Market concentration patterns
- Bank type distributions

Non-Comparable Aspects:

- **Coverage:** Mexico (L/C only) vs. others (all TF credit)
- **Granularity:** Brazil lacks bank-level data
- **Time periods:** Different available ranges per country
- **Accounting systems:** Chile 2022 break requires caution

Key Insight: Cross-country comparisons valid for structural patterns (ratios, concentrations) but not absolute volumes.

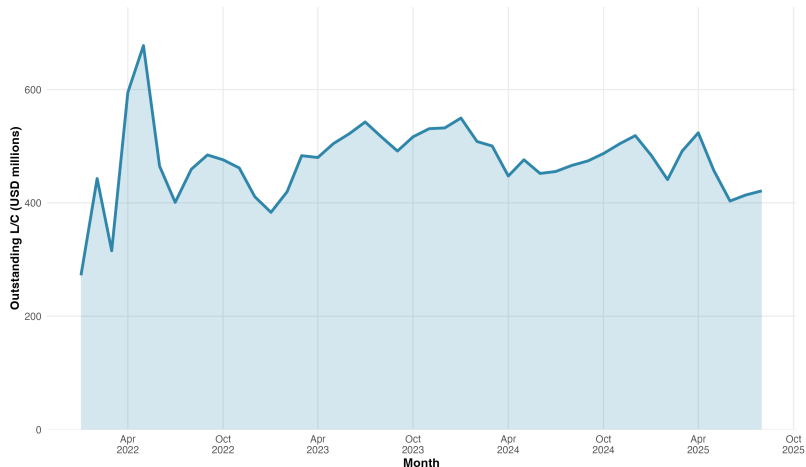
MEXICO

Trade Finance Analysis

6 Graphs | 2022-2025 | CNBV Reporte A-1219

Letters of Credit (Import L/C $\leq$ 180 days)

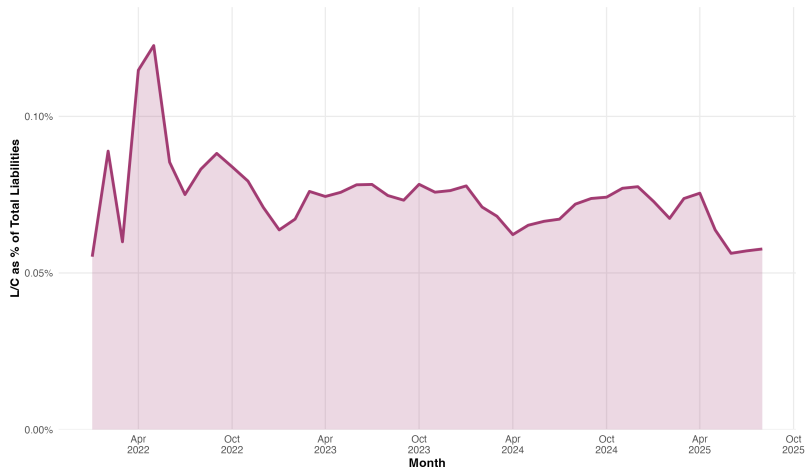
MEX-1: Outstanding Letters of Credit (Monthly Evolution)



Description: Monthly time series of total outstanding letters of credit issued by Mexican banks in USD millions.

Source: CNBV Reporte A-1219

MEX-2: Letters of Credit as % of Total Liabilities

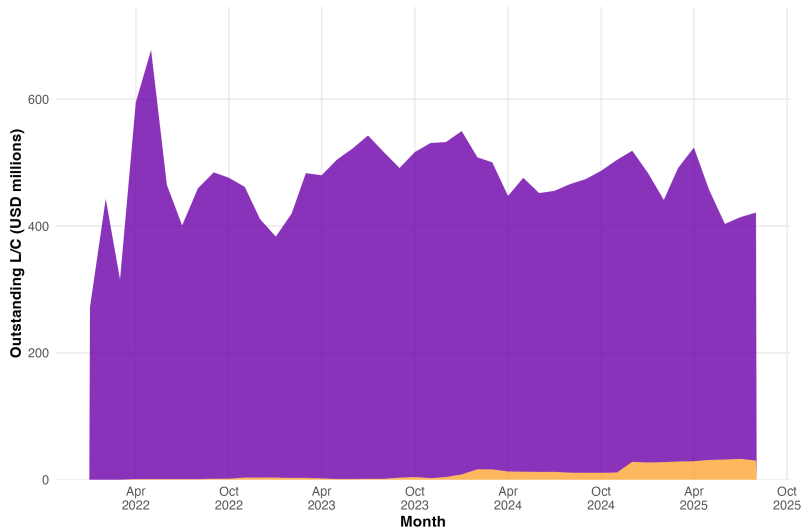


Description: Letters of credit as percentage of total bank liabilities. Monthly frequency 2022-2025.

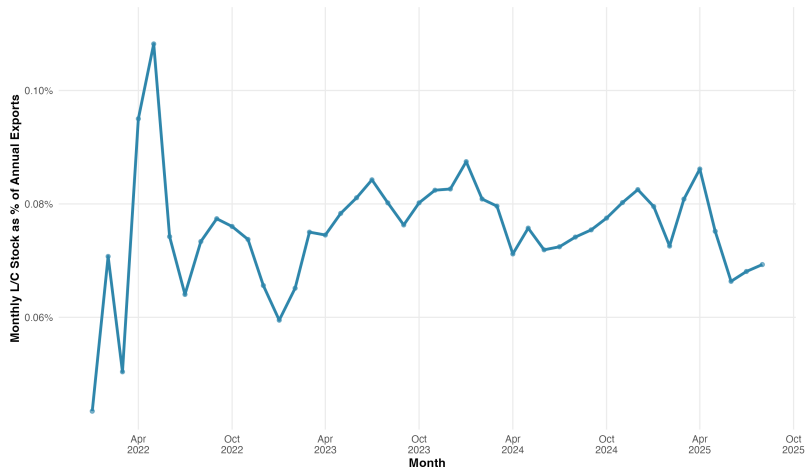
Source: CNBV Reporte A-1219

Notes: Range: 0.055% - 0.123%

MEX-3: Letters of Credit by Type of Bank



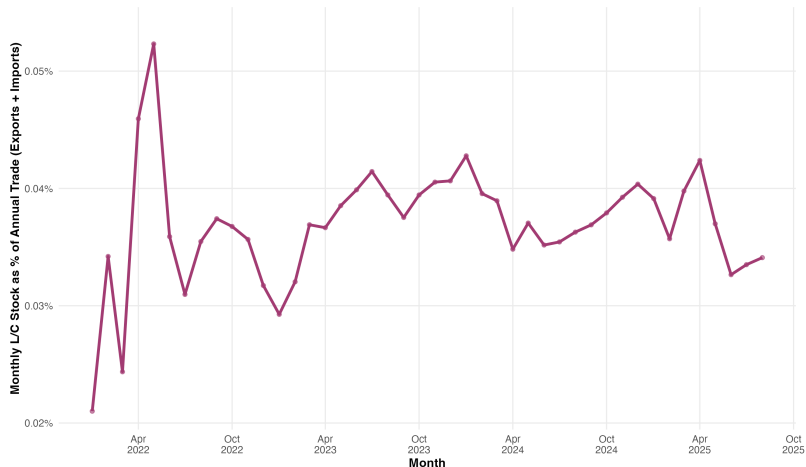
MEX-4: L/C as % of Annual Exports



Description: Monthly L/C stock as percentage of annual exports. Methodology: monthly stock / annual exports $\times$ 100.

Source: CNBV Reporte A-1219 + GMD

MEX-5: L/C as % of Annual Total Trade

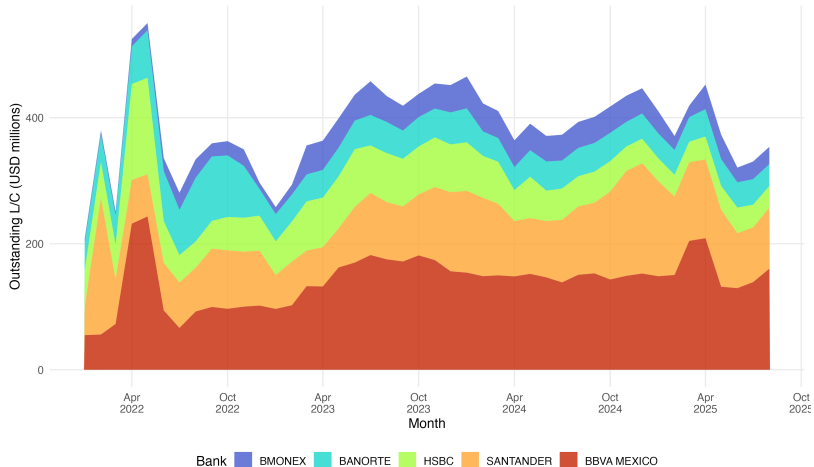


Description: Monthly L/C stock as percentage of annual total trade (exports + imports).

Methodology: $\text{monthly stock} / \text{annual (X+M)} \times 100$.

Source: CNBV Reporte A-1219 + GMD

MEX-6: L/C by Top 5 Banks (Monthly)



Description: Monthly stacked area showing L/C concentration among top 5 issuing banks. 2022-2025.

Source: CNBV Reporte A-1219

Notes: Top 5 banks represent majority of L/C market

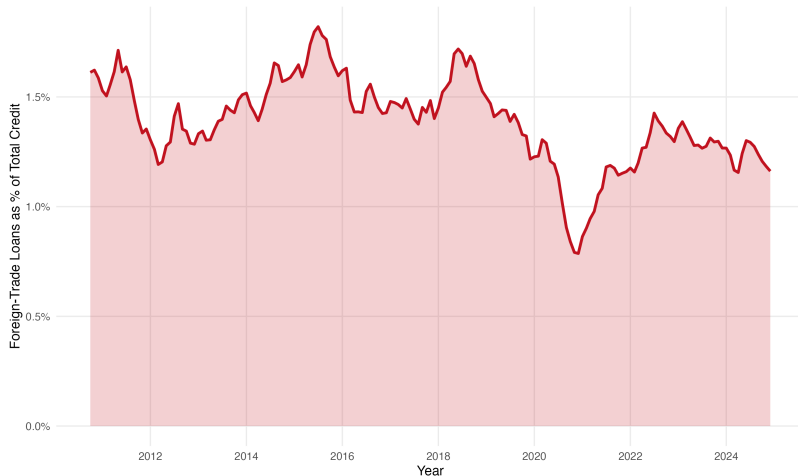
PERU

Trade Finance Analysis

7 Graphs | 2010-2024 | SBS Balance de Comprobación

Foreign-Trade Credit (Comercio Exterior)

PER-1: Foreign-Trade Credit as % of Total Credit

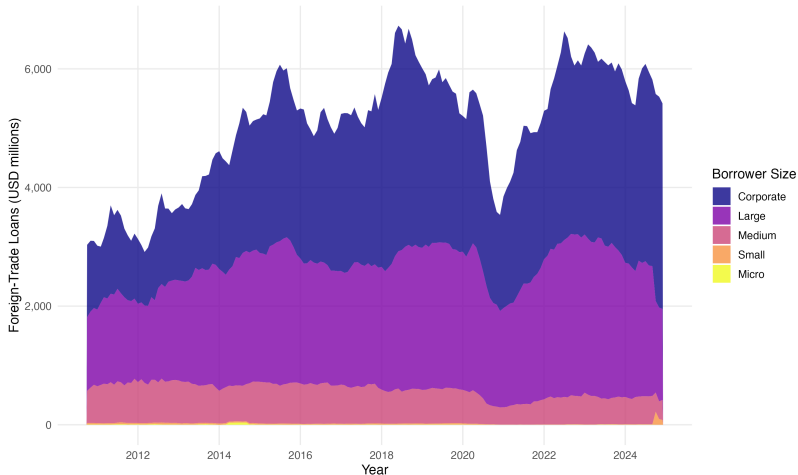


Description: Foreign-trade credit as share of total credit portfolio. Monthly 2010-2024.

Source: SBS Credit Registry

Notes: Down 2.7% of total credit

PER-2: Foreign-Trade Credit by Borrower Size

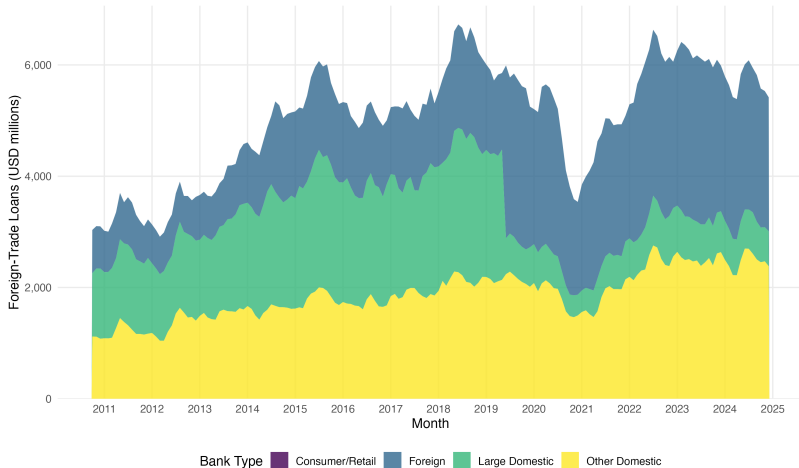


Description: Annual grouped bars showing TF distribution across borrower size categories. 2010-2024.

Source: SBS Balance de Comprobación

Notes: Financiamiento: Micro, Small, Medium, Large, Corporate

PER-3: Foreign-Trade Credit by Type of Bank (Monthly)

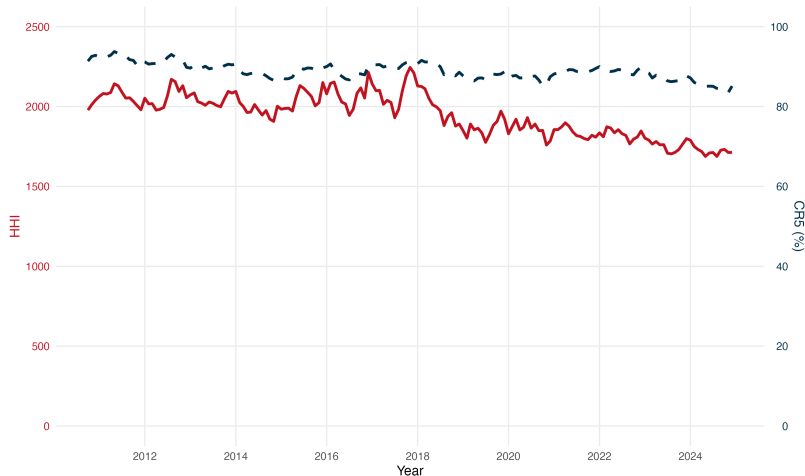


Description: Monthly stacked area chart showing TF composition by bank type. 2022-2024.

Source: SBS Balance de Comprobación

Notes: Foreign loans = 25-40% of total loans. Includes BBVA, Scotiabank, Citibank.

PER-4: Market Concentration (HHI)

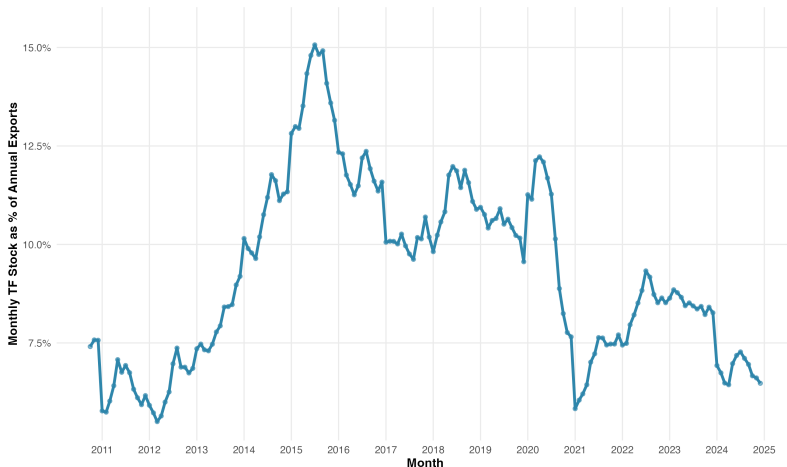


Description: Herfindahl-Hirschman Index tracking market concentration. 2010-2024.

Source: SBS Balance de Comprobación

Notes: Top 5: BBVA, BCP, Scotiabank, Interbank, Continental. CR5: 75.99%

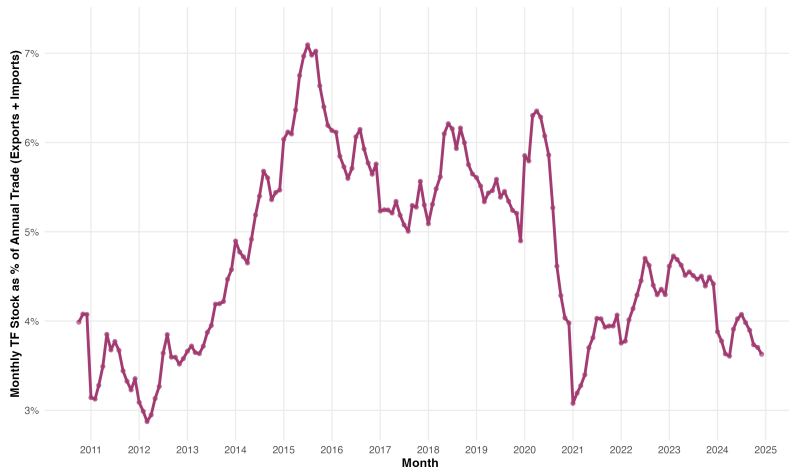
PER-5: TF as % of Annual Exports



Description: Monthly TF stock as percentage of annual exports. Methodology: $\text{monthly stock} / \text{annual exports} \times 100$.

Source: SBS Balance de Comprobación + GMD

PER-6: TF as % of Annual Total Trade

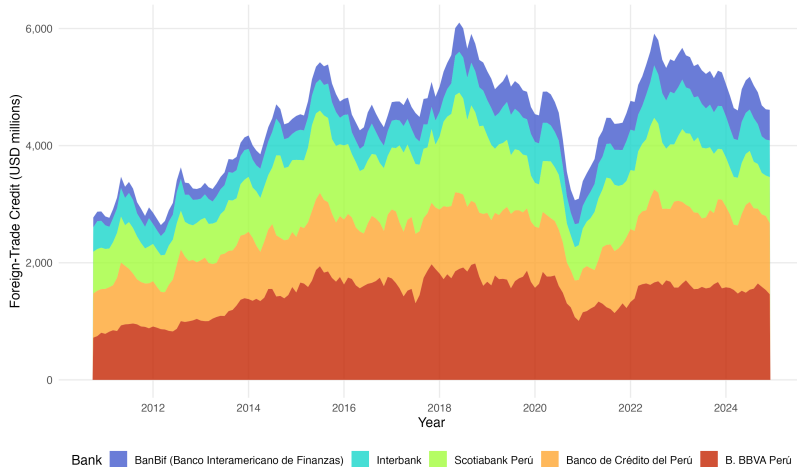


Description: Monthly TF stock as percentage of annual total trade (exports + imports).

Source: SBS Balance de Comprobación + GMD

Notes: Range: 2.0% - 7.1%

PER-7: TF by Top 5 Banks (Monthly)



Description: Monthly stacked area showing foreign-trade credit concentration among top 5 institutions. 2010-2024.

Source: SBS Balance de Comprobación

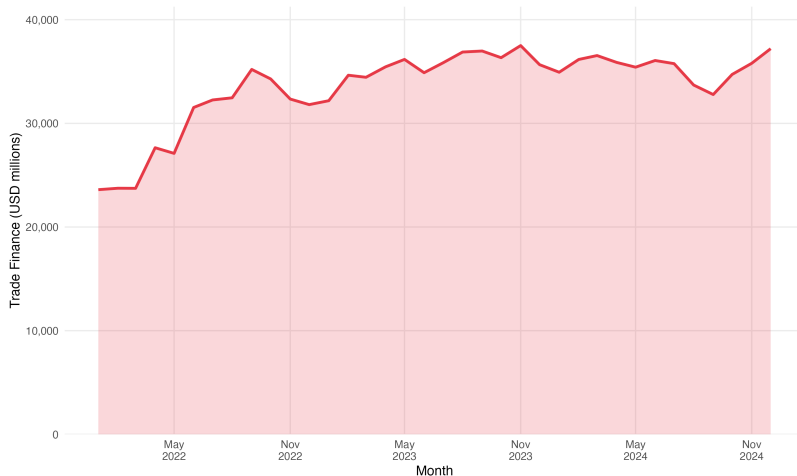
CHILE

Trade Finance Analysis

9 Graphs | 2022-2024 | CMF API

CMF/IFRS 9 Period (Accounting Break Jan 2022)

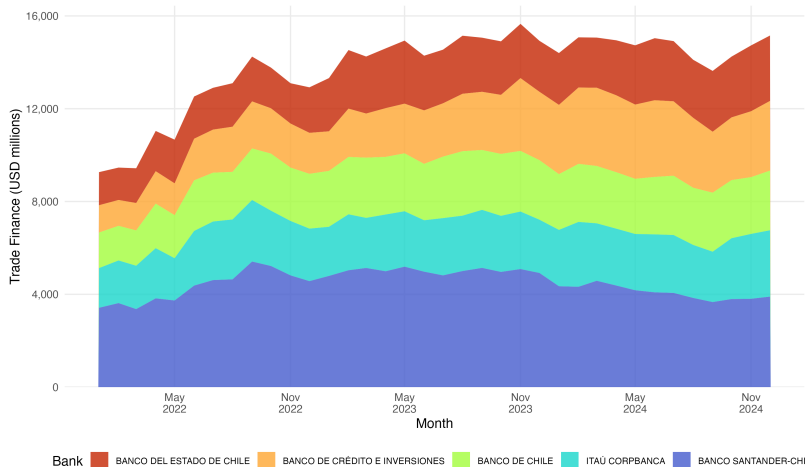
CHL-1: Foreign-Trade Loans as % of Total Loans



Description: TF loans as share of total loan portfolio. Series 2015-2024 with SBIF→CMF/IFRS9 accounting transition.

Source: CMF API

CHL-2: TF by Top 5 Banks (Monthly)

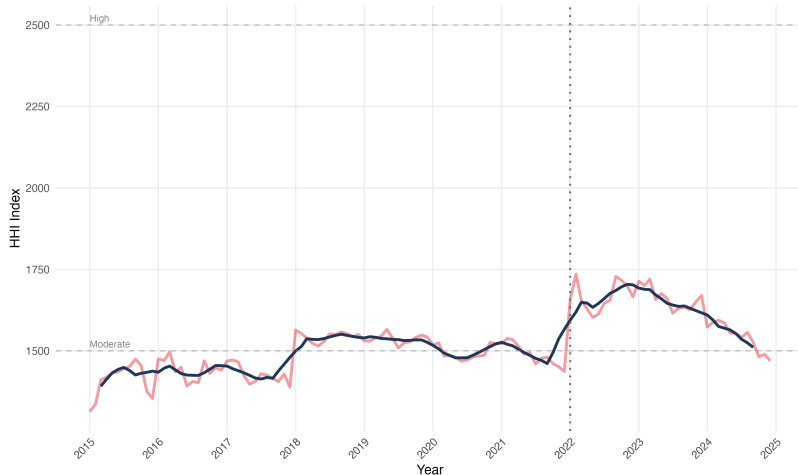


Description: Monthly stacked area showing TF distribution among top 5 banks. 2022-2024.

Source: CMF API

Notes: Top 5 banks account for majority of TF market

CHL-3: Market Concentration (HHI)

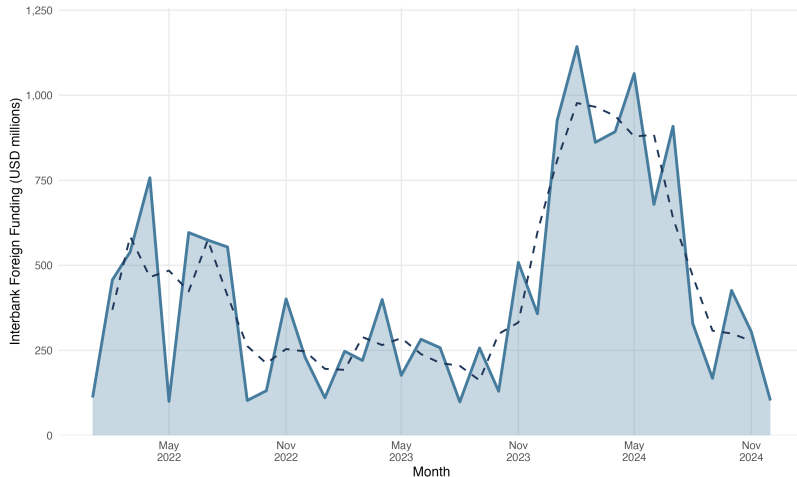


Description: Herfindahl-Hirschman Index tracking market concentration 2015-2024.

Source: CMF API

Notes: HHI < 1500 = moderate concentration

CHL-4: Interbank Foreign Funding



Description: Funding from foreign correspondent banks for Chilean trade operations. Monthly 2022-2024.

Source: CMF API

CHL-5: Letters of Credit as % of Liabilities

plots/chile/chl_05_lc_pct_liabilities.png

CHL-6: L/C by Top 5 Banks (Monthly)

`plots/chile/chl_06_lc_top5_banks.png`

CHL-7: TF as % of Annual Exports

plots/chile/chl_07_tf_pct_exports.png

CHL-8: TF as % of Annual Total Trade

plots/chile/chl_08_tf_pct_trade.png

CHL-9: Trade Finance Composition by Category

plots/chile/chl_09_tf_composition.png

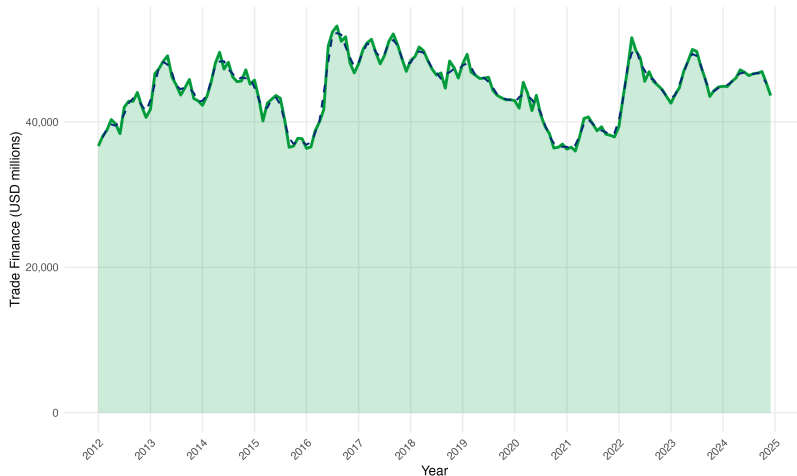
BRAZIL

Trade Finance Analysis

8 Graphs | 2012-2024 | BCB Monthly Reports

Foreign-Trade Credit (No Bank-Level Data)

BRA-1: Foreign-Trade Credit as % of Total Credit

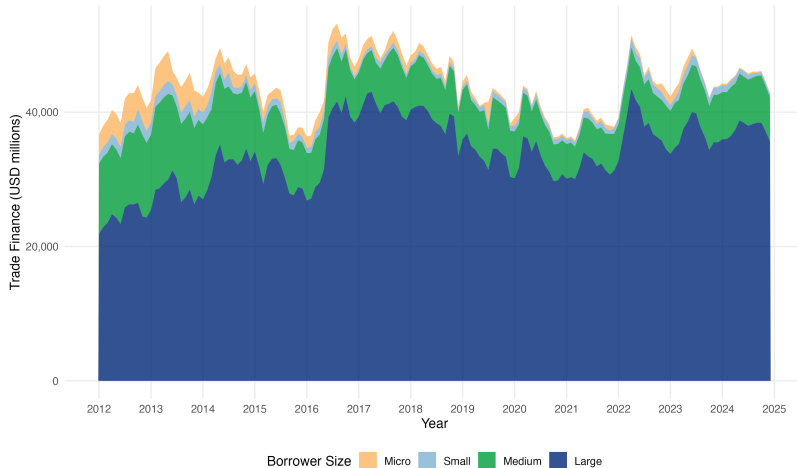


Description: TF credit as share of total credit. National aggregate. Monthly 2012-2024.

Source: BCB SCR

Notes: Represents 3.8% of total credit. Includes AGC, AGE, FINIMP, NCF, and others.

BRA-2: Foreign-Trade Credit by Borrower Size

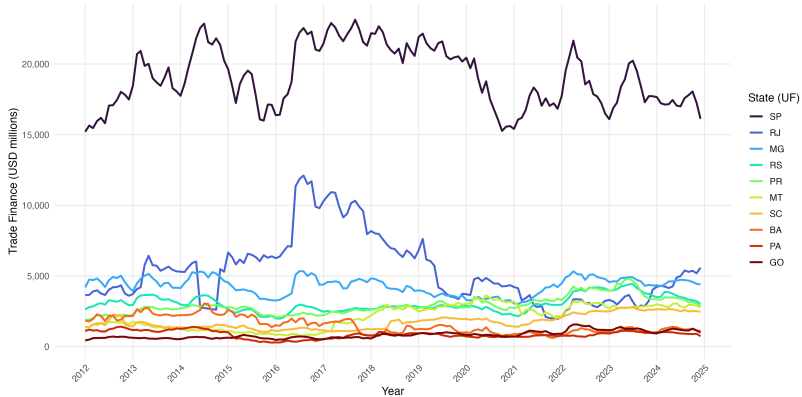


Description: Annual grouped bars showing TF distribution across enterprise size categories. 2012-2024.

Source: BCB Monthly Reports

Notes: Since BCB/BNDES/SEBRAE started in 2012, 54% of companies are in 'Large' category, 29% in 'Medium', 14% in 'Small', and 3% in 'Micro'.

BRA-3: Foreign-Trade Credit by Region

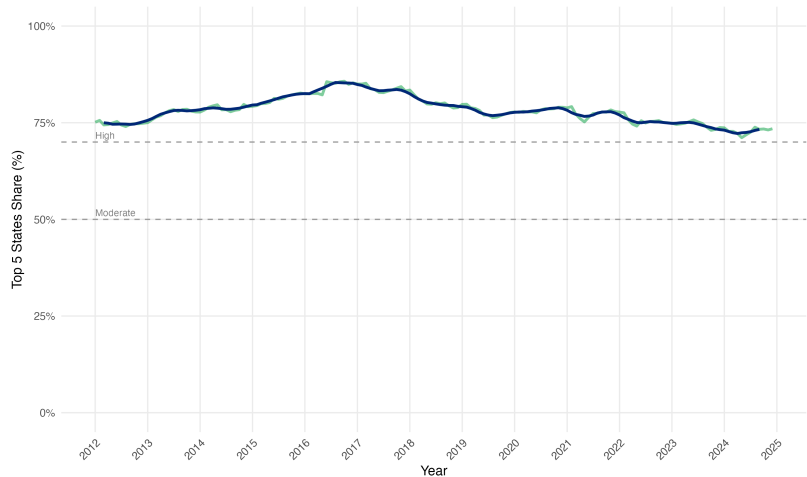


Description: Annual grouped bars showing TF geographic distribution across Brazilian regions. 2012-2024.

Source: BCB Monthly Reports

Notes: Southeast + South: 81% of TF. São Paulo: 40%. North: 2.2%.

BRA-4: Regional Concentration

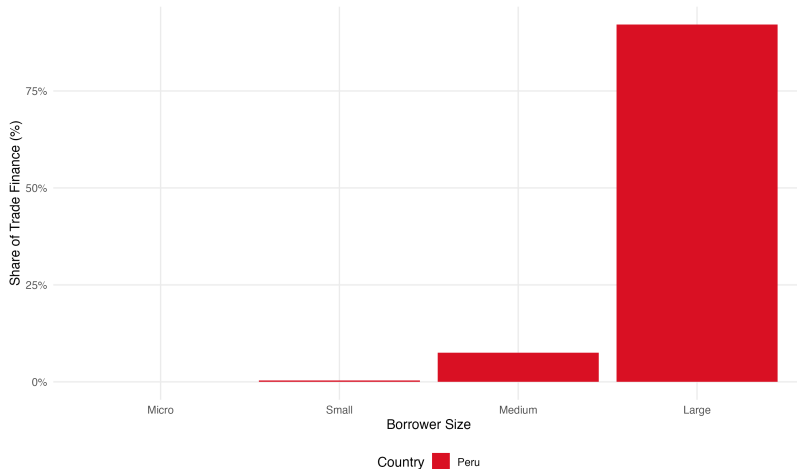


Description: Line chart showing regional concentration metrics evolution 2012-2024.

Source: BCB Monthly Reports

Notes: Top 5 states (SP, RJ, MG, PB, SC) 73% of national volume

BRA-5: Size Distribution Comparison: Brazil vs Peru

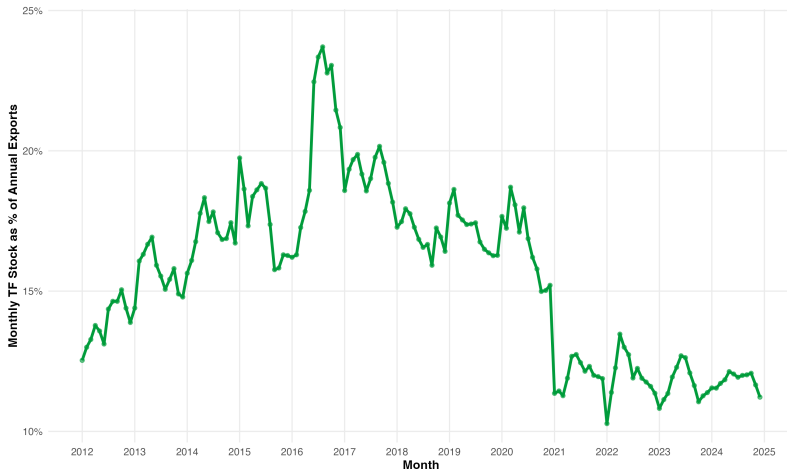


Description: Side-by-side grouped bars comparing TF borrower size distribution. 2023-2024 average.

Source: BCB Monthly Reports + SBS Balance de Comprobación

Notes: Brazil: 54%, Peru: 45%

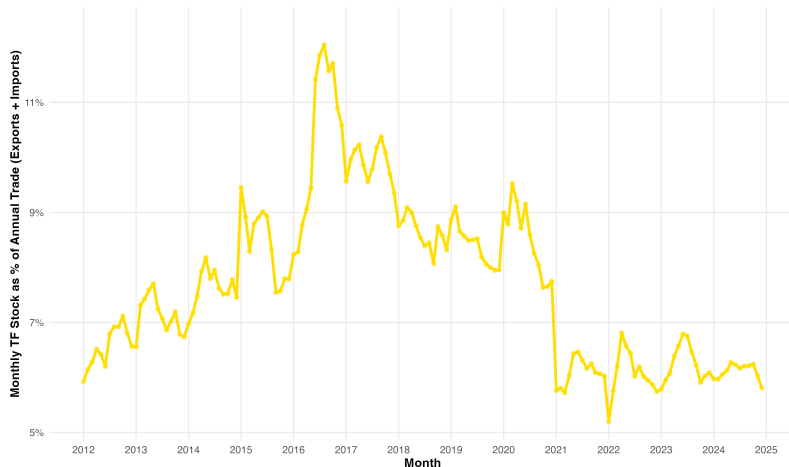
BRA-6: TF as % of Annual Exports



Description: Monthly TF stock as percentage of annual exports. Methodology: $\text{monthly stock} / \text{annual exports} \times 100$.

Source: BCB Monthly Reports + GMD

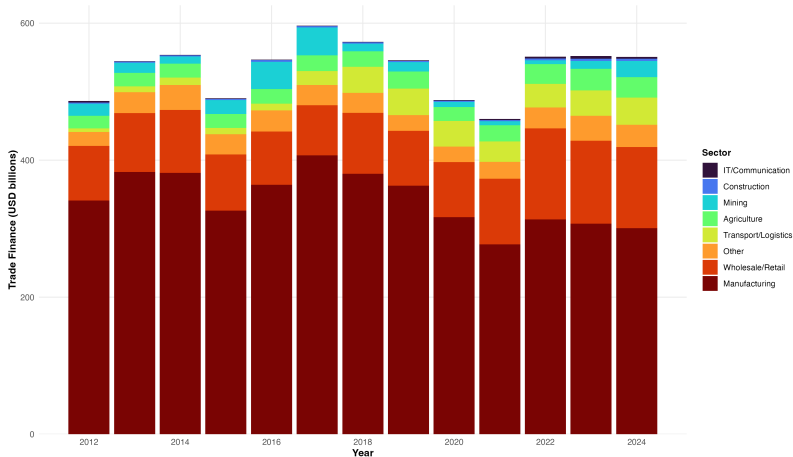
BRA-7: TF as % of Annual Total Trade



Description: Monthly TF stock as percentage of annual total trade. Methodology: $\text{monthly stock} / \text{annual (X+M)} \times 100$.

Source: BCB Monthly Reports + GMD

BRA-8: Trade Finance by Economic Sector



Description: Annual stacked bar showing TF distribution across top 10 economic sectors. 2012-2024.

Source: BCB Monthly Reports

Notes: Top 3: Manufacturing (62.9%), Wholesale/Retail (17.0%), Other (5.3%).

End of Presentation

Country Trade Finance Analysis
November 2025

All graphs available in `plots/` directory
Complete captions in `captions/captions.csv`